RUTGERS

Semantics I

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Please pick up your **name card** from the front

office hours: Mondays, 1:00-2:00 pm

Basement, Room 003, **Linguistics Building**

Midterm: Morphological Analysis, Hebrew

(a) [lavan] ‘white’

(b) [kaxol] ‘blue’

(c) [jaɣok] ‘green’

(d) [tsahov] ‘yellow’

(e) [vaɣod] ‘pink’

(f) [ʃaxoɣ] ‘black’

(g) [lvanvan] ‘whitish’

(h) [kxalxal] ‘bluish’

(i) [jɣakɣak] ‘greenish’

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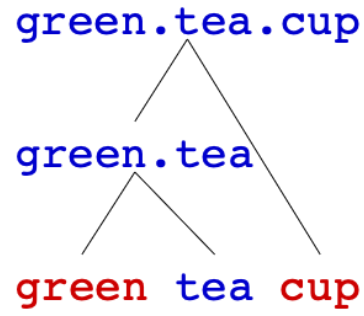
never be arrogant towards other people's languages

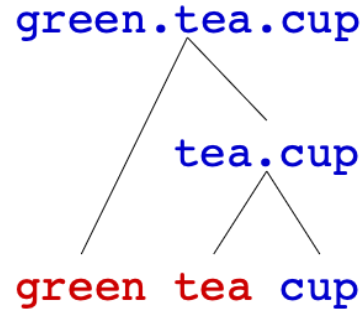
Meaning in Language

- so far this semester, we've focused on the form or structure of language

- **Morphology**: study of the structure of words
  - morphemes, morphological processes, morphological trees

- **Syntax**: study of the structure of sentences
  - parts of speech, constituents, phrase structure rules, syntax trees



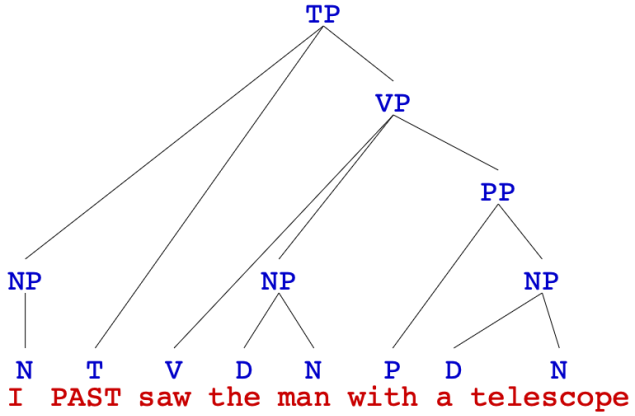


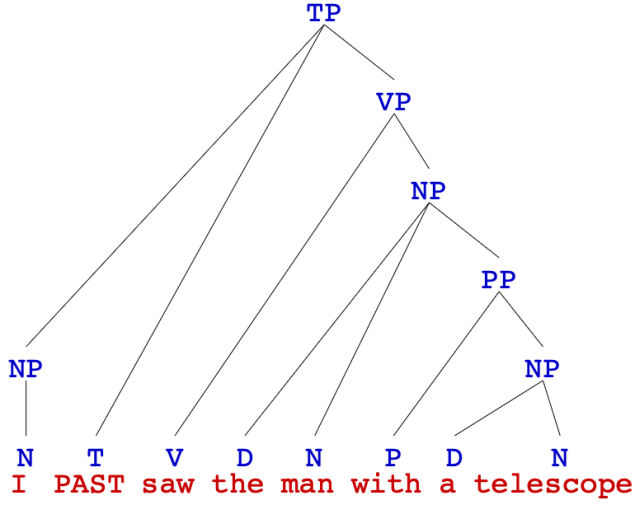
Meaning in Language

of course, language is **NOT** just structure – language has **meaning** – and we've mentioned meaning on several occasions:

- **Morphology**
  - morphemes are the smallest linguistic unit with *meaning*
  - derivation changes a word's core *meaning*
  - order of affixation reflects a word's *meaning*

- **Syntax**
  - parts of speech can be defined based on their *meaning*
  - ambiguous sentences like '*The girl saw the bear with the telescope*' show us that syntactic structures contribute to the *meaning* of sentences





Meaning in Language

in this and the next unit, we're going to focus on **meaning** more specifically and we can differentiate three **types** of linguistic meaning:

literal meaning

- **Semantics** studies *literal meaning* in language

implied meaning

- **Pragmatics** studies *implied meaning* in language

social meaning

- **Sociolinguistics** and **Linguistic Anthropology** study *social meaning*

Literal Meaning and Implied Meaning

- as a system for communicating ideas, language must have **meaning**
- but meaning in language goes **beyond** the **literal**
    - imagine the following conversation:

      A: *Do you want to go out tonight*?
      B: *I have an exam in the morning*

    - what has B literally said?
    - what has B implied that they have not said explicitly?
- as in this example, much of what we say has different levels of **literal** and **implied** meaning



Social Meaning

- language also has **social meaning**, since it's a tool for **social interaction**:
  - we use **small talk** to establish and reinforce social bonds with others
  - the way we speak can indicate our attitude toward others (friendliness, solidarity, irritation…)
- we actively **choose** to speak a particular way to convey a particular social identity
  - good morning to a professor
  - 'sup to a friend
  - do we usually do it other way around?
  - homosexual voice? [just an example, not sterotype]
- we often unconsciously **accommodate to the speech** of people we like, adopting phrases and speech patterns from them without even noticing it
  - have you noticed your speech changing since you came to college?

Social Meaning

language **varies** by region, community, and other social factors

- so the way we speak often **conveys information** like where we're from, our age, gender, etc.
- we often strongly **identify** with the language and dialect we grew up speaking
- we can actively use language to **signal our membership** in certain communities
    - A: *Did you eat my cake*?
      B: *I **never** ate your cake*.



Semantics



- in this unit, we'll focus on **Semantics** – the literal meaning of language

- we can identify two levels of semantics: **lexical semantics** and **compositional semantics**

- **lexical semantics** – the meaning of words
  - lexicon refers to a speaker's mental vocabulary; words are known as lexical items, but lexical items can also include common multi-word expressions

- **compositional semantics** – the meaning of sentences
  - we said *A girl saw a bear with a telescope.* has two different meanings, and these meanings depend on syntax
  - meaning of a sentence = meaning of words + meaning of structure

Lexical Meaning

lexical semantics studies the **meaning of words** and lexical meaning is **conventional** and **arbitrary**

- lexical meaning is **conventional**

1. words mean what they mean because we agree as a speech community that's what they mean – that is, words are defined by **social convention**

2. consider words with slang meanings like *sick* 'amazing, excellent'

    - how did *sick* come to mean 'great'?

    - someone starting using it with that meaning, and it caught on – the rest of the community agreed it can be used that way, too

3. same thing with **technological terms** like *mouse*, *screen*, *tablet*, even *computer*:

    - all of these words originally had other meanings, but when we invented new technology and needed a name for it, we agreed these words can have new meanings

Lexical Meaning

lexical semantics studies the **meaning of words** and lexical meaning is **conventional** and **arbitrary**

- lexical meaning is **arbitrary**
  - there is no inherent connection between the sound of a word and its meaning

- we know this because:
  - same **sounds** can have different **meanings** in different languages:
    Spanish *taco* 'taco' vs. **Japanese** *tako* 'octopus'
    English *eat* 'eat' vs. **Kazakh** *it* 'dog'

  - same **meaning** can be conveyed with different **sounds** in different languages:
    English *dog* vs. **Spanish** *perro*

  - meaning and sound of words can **change over time**:
    nice 'foolish' > 'kind', *knight* [knixt] > [najt]

- main exception: **onomatopoeia** – words that mimic sounds
  - even **onomatopoeia** can be a bit arbitrary: cockadoodledoo, quiquiriquí, kokekokko, wowo

Sense and Reference

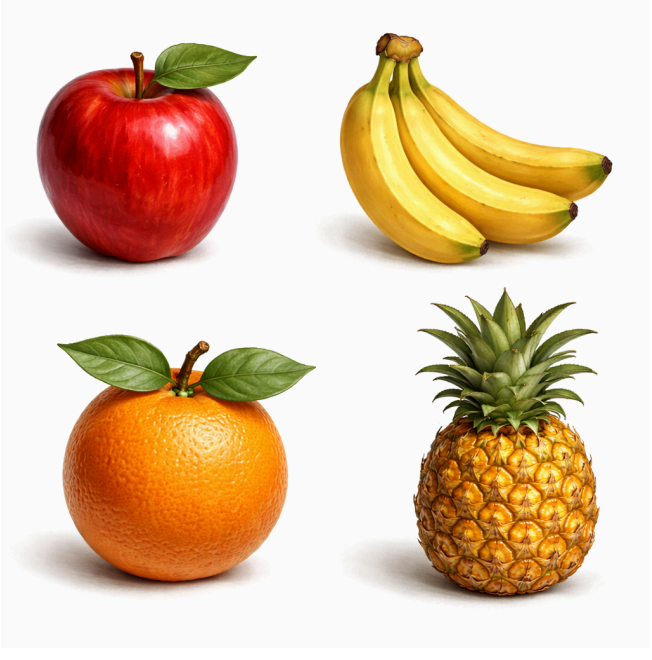
- so we know that words have meaning: **lexical semantics**
- and we know that lexical meaning is **conventional** and **arbitrary**
- but how do we specify what a word means?
- think of the words **fruit** and **animal**:
  - what image comes to mind when you think of a **fruit** or an **animal**?
  - how might you try to define these words?





Sense and Reference

a test: which group is **animal** and which group is **fruit**? how do you know?





Sense and Reference

a test: how about these new thigs? **animals or fruit**? how do you know?







Sense and Reference





there are two ways of thinking about a word's meaning:

sense and **reference**

- the **sense** of a word is:
  - the mental concept of a word's meaning
  - a description of this mental concept: dictionary definition

- for instance, the sense of fruit might be:
  - a seed-bearing structure in flowering plants

Sense and Reference





there are two ways of thinking about a word's meaning:

sense and **reference**

- the **reference** of a word is:
  - the set of things in the world that the word refers to
  - each thing the word refers to is a **referent** of the word

- the reference of the word fruit would include:
  - the apple I had for breakfast
  - the bananas they have at Cafe West
  - the oranges they sell at Costco...

Sense and Reference





- defining words by **reference** helps us understand how we can categorise new items we come across:
  - by comparing them to the referents we already know

- it's also more in line with how we **learn** words:
  - we don't learn what an animal is by learning that it's a "*multicellular eukaryotic organism [in] Animalia*" (Wikipedia)

  - we learn what an animal is by learning that a **puppy** is an animal and a **kitty** is an animal and a **froggy** is an animal...

  - we learn what a fruit is by observing what our caregivers call fruit

Sense and Reference





in this class, we'll define words by **reference**, which is the **set** of things in the world (referents) that the word refers to

- once we define a word as a **set** or collection of things, we can also **compare** the sets of things that different words refer to

1. some words refer to **entirely different** sets of things
2. sometimes there is overlap between **two sets** of referents

is there any reference overlap in each pair below?

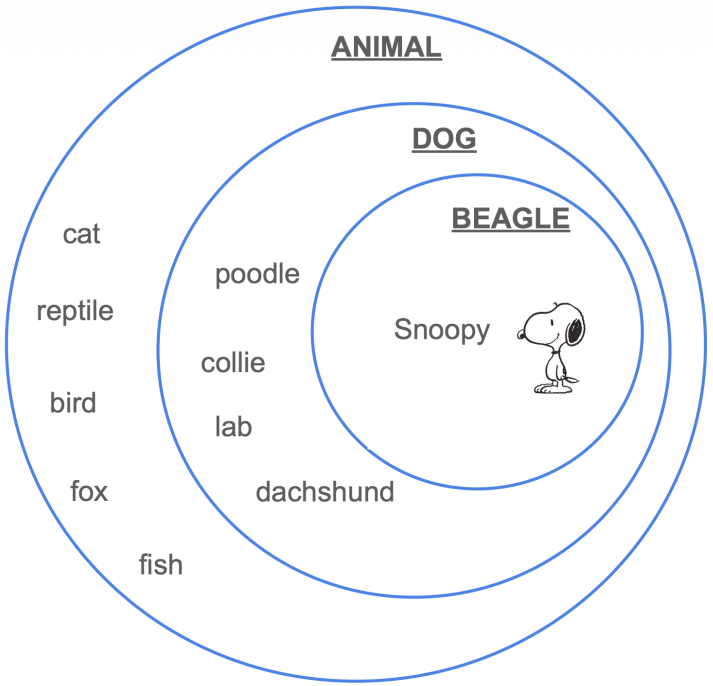
- dog vs. animal
- dog vs. canine
- dog vs. husky

Semantic Relations

- **synonym:** words that have the same reference / refer to the same set of things in the world
    - **drinks** and **beverages**
- **hypernym:** words whose reference is a **superset** of another words reference
    - **student** is a hypernym to **linguistics student**
- **hyponym:** words whose reference is a **subset** of another words reference
    - **dog** is a hyponym to **mammal**
- **antonym:** words that have opposite meanings



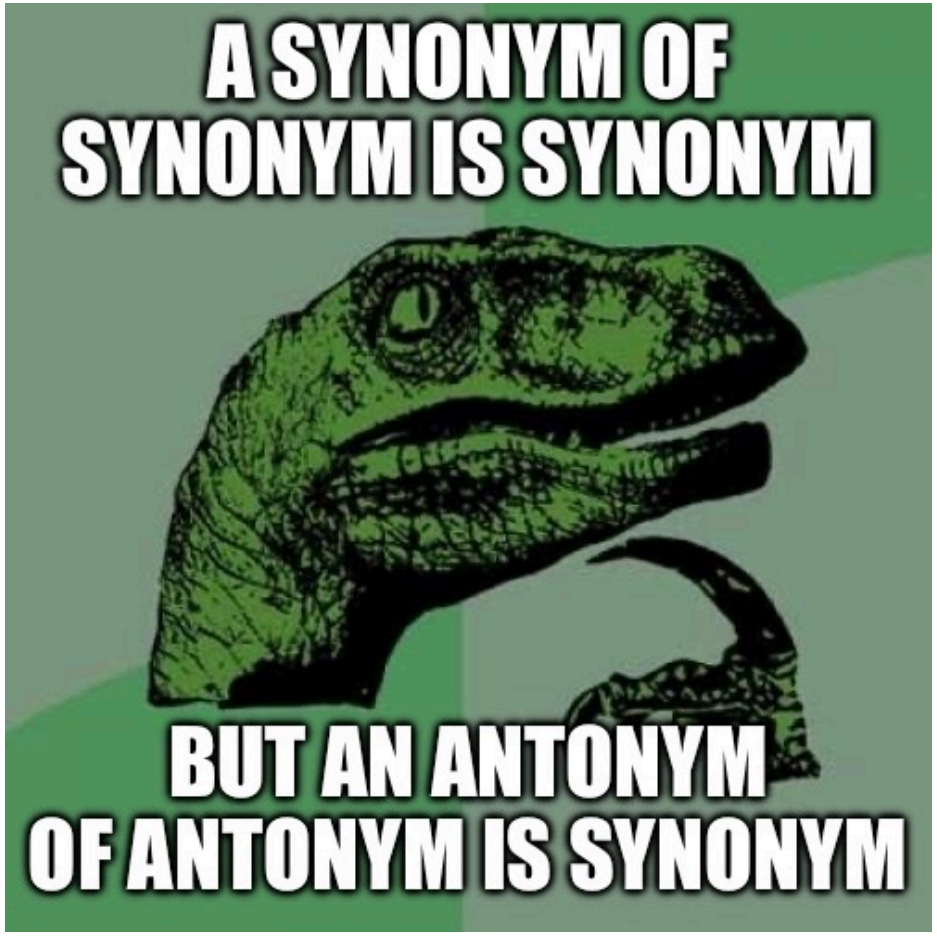
[[drink]] = [[beverages]]



Semantic Relations: Antonyms

antonym: words that have opposite meanings, and it turns out to be difficult to define in terms of reference

- **complementary pairs:** items that can't be in both categories
  - **alive** vs. **dead**
- **gradable pairs:** opposite ends of a scale
  - **young** vs. **old**
- **reverses:** opposite directions
  - **up** vs. **down**
- **converses:** same action, different perspective
  - **buy** vs. **sell**



Exercise I: Semantic Relations

- are the following word pairs **synonyms**, **antonyms**, or a **hypernym-hyponym pair**?
- if the last, which is the hypernym and which is the hyponym?



1. bird – avian
2. bird – sparrow
3. bird – animal
4. synonym - antonym
5. hypernym - hyponym
6. Rutgers - universities
7. Linguistics - Mathematics
8. **MORE**?

Compositional Semantics

- so we've seen some ideas about **lexical semantics** – the meaning of words (or lexical items)

- beyond words, phrases and sentence can also have meaning - **compositional semantics**

- **COMPOSITIONALITY in semantics**:

    - the meaning of a **complex** expression is due to

    1. the meaning of its parts (i.e., lexical meaning);
    2. how these parts are **combined** (i.e., syntactic structure)

- compare these two sentences: (a) *It is snowing outside*. (b) *It is sunny outside*.

- both sentences are formed by *it is ... outside* + a *weather word*
- yet they cannot be true at the same time, and which one is true depends on *snowing* and *sunny*

Truth Values

- part of a sentence's meaning is its **truth value**: is it **TRUE** or **FALSE**

- What is the **truth value** of the following sentences?

    1. Earth is a planet.

    2. Fish are birds.

    3. Mars is in the Solar System.

    4. Cats are animals.



P	Q	R	$P \to Q$	$Q \to R$	$(P \to Q) \wedge (Q \to R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

Complex Truth Values

- some sentences make multiple propositions, each of which has its own truth value:

- *Earth is a rocky planet*.
  - proposition 1: Earth is a planet. **TRUE**
  - proposition 2: Earth is rocky. **TRUE**

- *Merlin speaks Kazakh and Greek*.
  - proposition 1: Merlin speaks Kazakh. **TRUE**
  - proposition 2: Merlin speaks Greek. **FALSE**



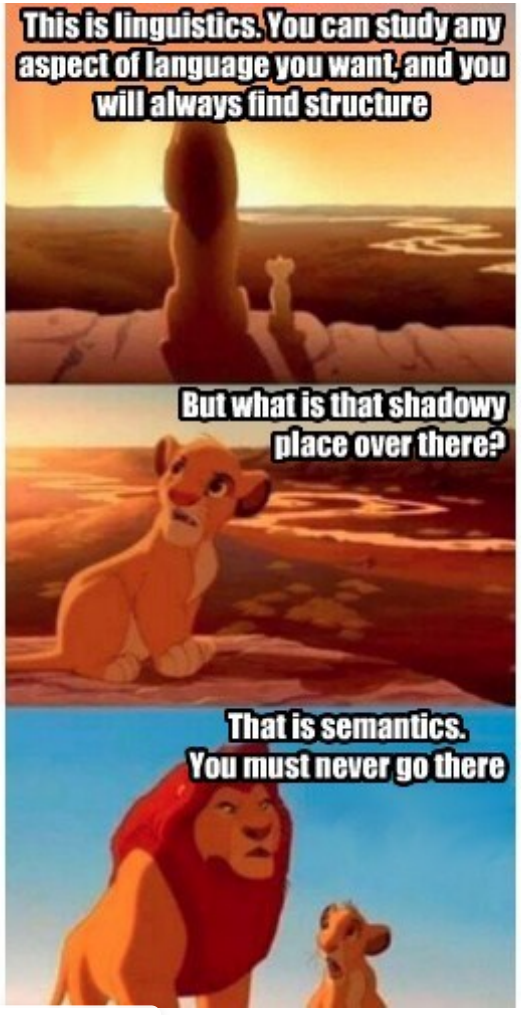
Complex Truth Values

- some sentences make multiple propositions, each of which has its own truth value:
- the truth value of a sentence **depends on** the truth value of its propositions
- all of the propositions in a sentence must be **TRUE** for the entire sentence to be **TRUE**

- *Earth is a rocky planet*.
  - proposition 1: Earth is a planet. **TRUE**
  - proposition 2: Earth is rocky. **TRUE**
  - sentence truth value: **TRUE**
- *Merlin speaks Kazakh and Greek*.
  - proposition 1: Merlin speaks Kazakh. **TRUE**
  - proposition 2: Merlin speaks Greek. **FALSE**
  - sentence truth value: **FALSE**



Complex Truth Values



- a sentence might also give rise to other, **unspoken propositions** – often related through synonyms, hypernyms, and hyponyms
- the sentence *I have a dog.* *gives rise to other propositions:*
  - *I have a pet.*
  - *My pet is a canine.*
  - *My pet is an animal.*
- *all of these propositions must also be* ***TRUE*** *for the sentence to be* ***TRUE****:*
  - *If my pet is an animal is false, then I have a dog cannot be true.*

Assignment IV will be published this Wednesday (**Oct 21**)
and is due next Friday **Oct 30** at **11:59 pm**

continue reading: the chapter **Semantics I: Compositionality**
from Victoria Fromkin et al.'s *An Introduction to Language*



Slides created via the R package **xaringan**.